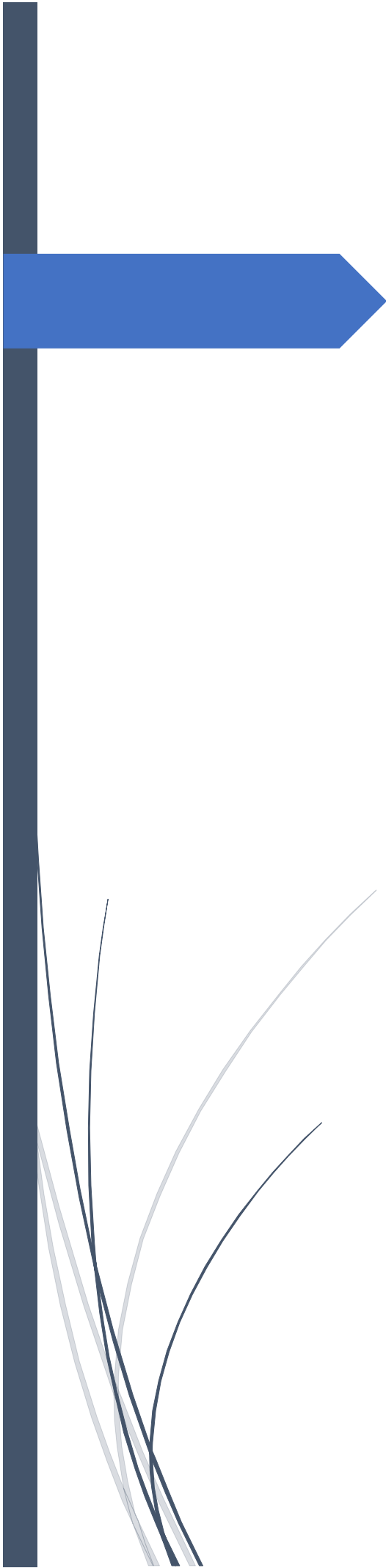




Employee Retention Analysis Report

Employee Attrition Prediction Using Machine Learning

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Title: Employee Attrition Prediction Using Machine Learning

➤ **Executive Summary:**

Employee attrition is a significant challenge for organizations as it affects productivity, increases recruitment costs, and impacts overall business performance. This project applies Data Science and Machine Learning techniques to analyze employee attrition patterns using the IBM HR Analytics dataset. A predictive model was developed to identify employees who are at risk of leaving the organization, enabling HR teams to make proactive, data-driven retention decisions.

➤ **Project Objective:**

The objective of this project is to analyze employee-related data, identify factors contributing to employee attrition, and build a predictive machine learning model capable of accurately forecasting employee turnover. The project demonstrates how HR analytics can be leveraged to improve employee retention strategies and organizational decision-making.

➤ **Dataset Overview:**

- Dataset: IBM HR Analytics Employee Attrition Dataset
- Total Records: 1,470 Employees
- Total Features: 35
- Target Variable: Attrition (Yes/No)

The dataset includes employee demographics, job characteristics, salary information, work environment, performance metrics, and other HR-related attributes used for predictive analysis.

➤ **Data Preprocessing:**

To ensure data quality and model performance, the following preprocessing steps were performed:

- Imported and explored the HR dataset.
- Performed data quality assessment.
- Verified the absence of missing values.
- Converted the target variable into numerical format.
- Applied Label Encoding to categorical features.
- Selected predictor variables and target variable.
- Split the dataset into training and testing datasets using an 80:20 ratio.

➤ **Exploratory Data Analysis:**

Exploratory Data Analysis (EDA) was conducted to understand employee behaviour and identify important trends.

The following visualizations were generated:

- Employee Attrition Distribution
- Age Distribution
- Gender Distribution
- Department Distribution
- Attrition by Gender

These analyses provided valuable insights into workforce characteristics and attrition patterns.

➤ **Machine Learning Model:**

A Gradient Boosting Classifier was selected due to its strong predictive performance for structured datasets.

- Model Configuration
- Algorithm: Gradient Boosting Classifier
- Training Dataset: 80%
- Testing Dataset: 20%
- Random State: 42

The model was trained using the processed dataset and evaluated using standard classification metrics.

➤ **Model Evaluation:**

The developed model achieved the following performance:

- Accuracy: 89%
- Precision, Recall, and F1-Score were calculated using the Classification Report.
- Model performance was further evaluated using a Confusion Matrix.
- The evaluation demonstrates that the model effectively predicts employee attrition with a high level of accuracy.

➤ **Business Insights:**

The analysis indicates that employee attrition is influenced by multiple demographic and workplace factors. Predictive analytics enables organizations to identify employees with a higher probability of leaving and implement proactive retention strategies before attrition occurs.

This approach supports evidence-based Human Resource management and improves workforce planning.

➤ **Recommendations:**

Based on the analytical findings, organizations should:

- Strengthen employee engagement initiatives.
- Improve work-life balance by monitoring overtime.
- Enhance career development and promotion opportunities.
- Increase employee satisfaction through targeted HR policies.
- Integrate predictive analytics into workforce planning and retention programs.

➤ **Conclusion:**

This project successfully demonstrates the practical application of Data Science, Machine Learning, and HR Analytics for solving a real-world business problem. By combining data preprocessing, exploratory data analysis, predictive modeling, and performance evaluation, an employee attrition prediction model with 89% accuracy was developed.

The proposed solution provides HR professionals with actionable insights to identify attrition risks, optimize retention strategies, and support data-driven decision-making. This project highlights the growing importance of predictive analytics in modern Human Resource Management and showcases the value of Machine Learning in improving organizational effectiveness and employee retention.